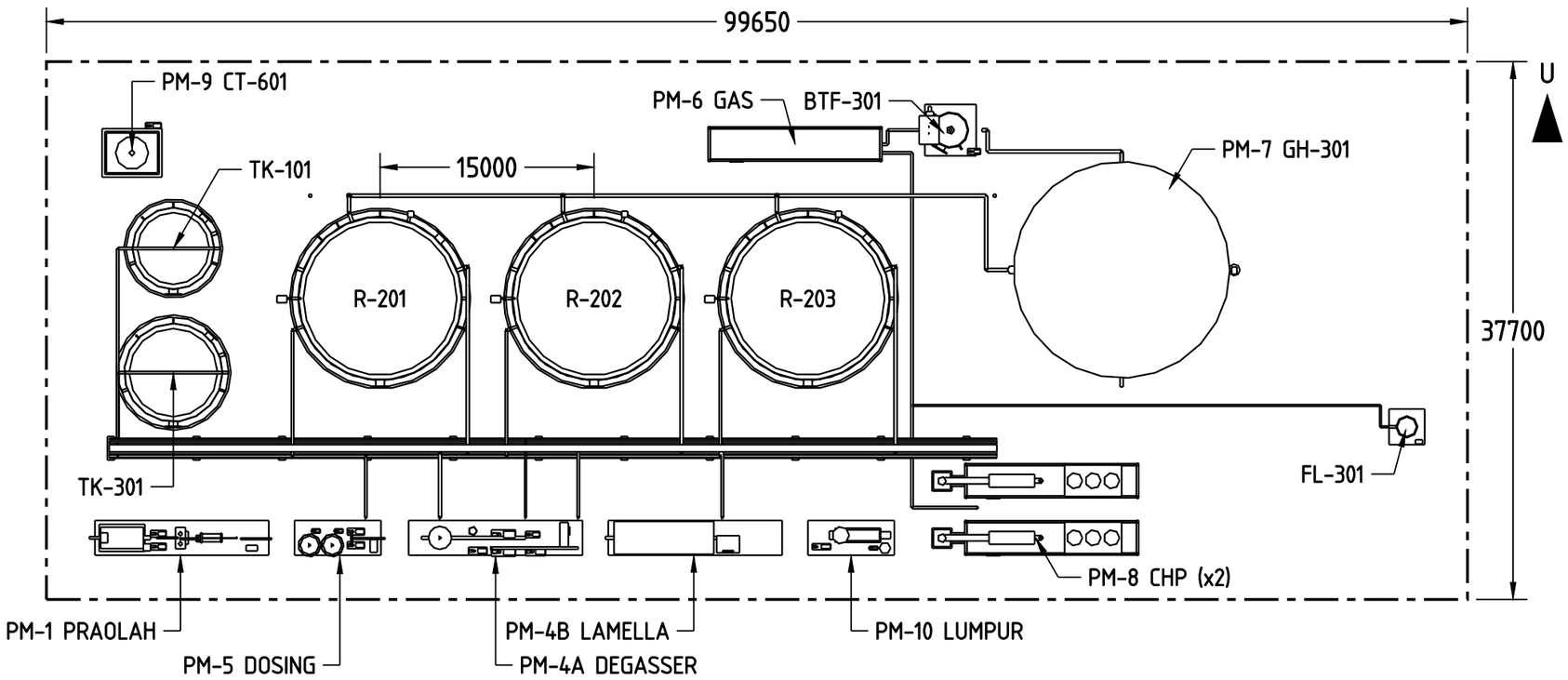
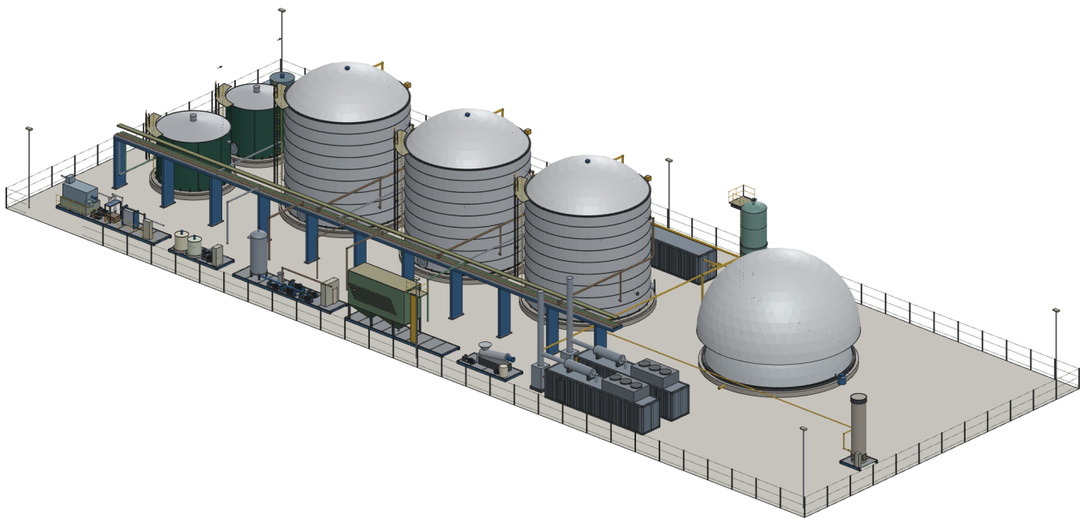


DENAH TATA LETAK (DARI MODEL 3D)

SKALA 1:400



ISOMETRI SUSUNAN PABRIK (TANPA SKALA)

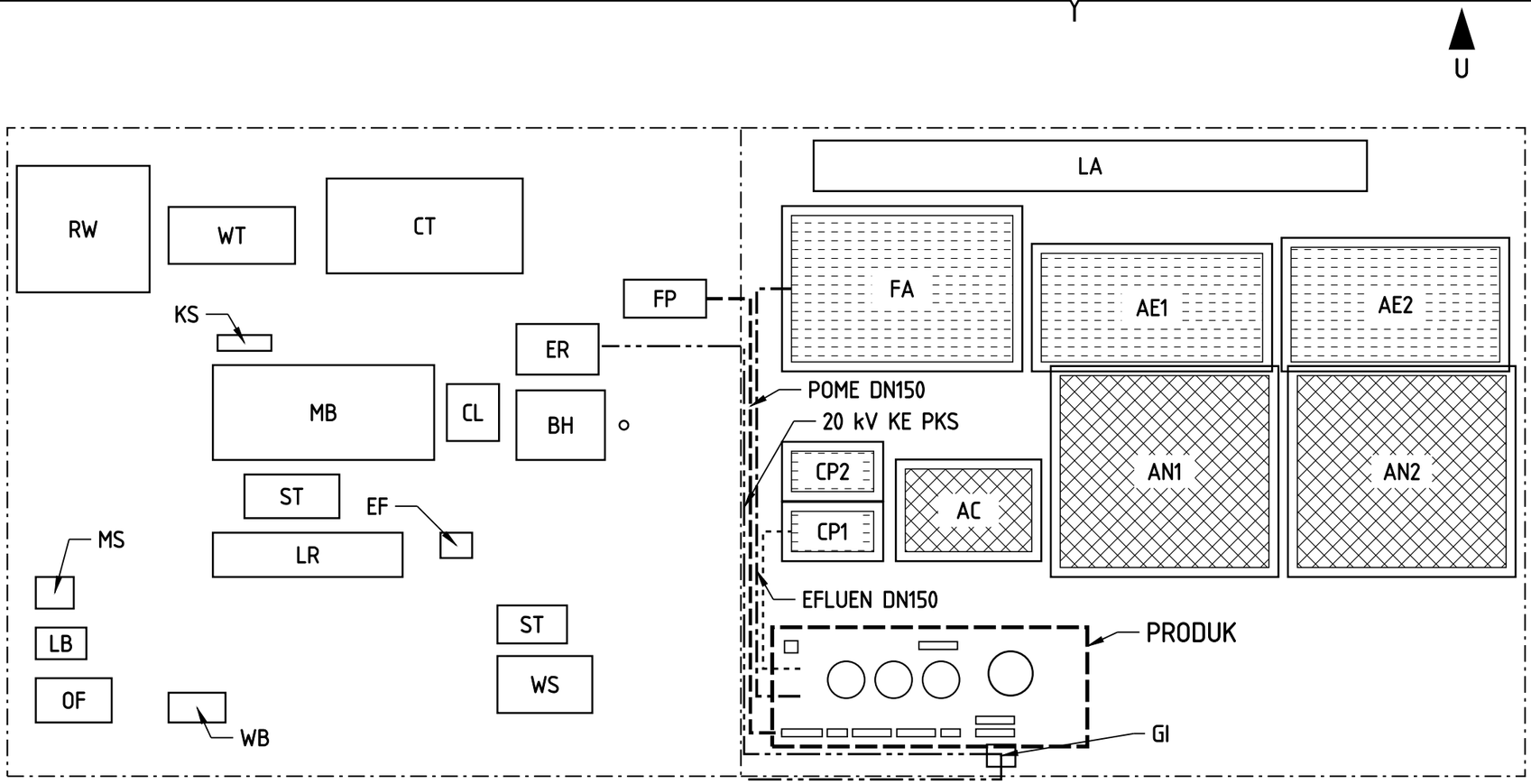
SET MODUL (DAPAT DIPINDAH)				
Modul	Nama	Jml	Pengiriman	Massa t
PM-1	Skid praolah POME	1	kontainer 40 ft (skid)	12.0
PM-2	Tangki GFS penyangga umpan	1	panel GFS dalam kontainer 40 ft	10.8
PM-3	Reaktor kontak termofilik (GFS)	3	panel GFS dalam kontainer 40 ft	113.8
PM-4	Skid kontak: degasser + lamella	1	2 skid 40 ft	18.0
PM-5	Skid dosing dan pompa umpan	1	skid 20 ft	8.0
PM-6	Skid pengolahan biogas	1	kontainer 40 ft + bejana BTF	14.0
PM-7	Penampung gas + flare	1	kontainer 20 ft + skid flare	7.0
PM-8	Kontainer CHP	2	kontainer 40 ft HC	40.0
PM-9	Skid pemulih panas + menara pendingin	1	skid 20 ft + sel CT modular	9.0
PM-10	Tangki GFS efluen + skid lumpur	1	panel GFS + skid 20 ft	21.3

Total 21 muatan kontainer/truk, ± 254 t. Plot 99.7 x 37.7 m = 0.38 ha (termasuk jalan 3 m).

SKALA SET (TANGKI STANDAR SAMA)					
PKS	Tangki GFS	Genset	Neto kWe	GWh/thn	Lamella m²
30 t/jam	2	2 x 400 kWe	681	3.5	54
45 t/jam	3	2 x 600 kWe	1025	5.3	81
60 t/jam	4	2 x 800 kWe	1369	7.0	108

hari nameplate; GWh/thn = rata-rata tahunan (util 0,75).

University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: General Arrangement			Rev: 2
	Author : NAMA TIM	Tata Letak Umum Set Biogas PKS 45 t TBS/jam (Rev 2)	Unit: mm	Scale: 1 : 400	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-GA-001			Sheet number: 1 / 9



INTERKONEKSI		
Rute (model 3D)	m	USD
POME DN150 fat pit -> PM-1	161	22565
Efluen DN150 -> kolam polishing	154	10755
Bypass darurat -> kolam pendingin	64	4492
20 kV gardu -> jaringan PLN	88	3938
20 kV ke rumah mesin PKS + trafo	256	28232
Total (masuk CAPEX)		69982

LAHAN	
Luas lahan	m²
Kolam POME sebelum	22361
Kolam tetap (polishing)	11281
Kolam dibebaskan	11081
Plot produk	3757

PEMERIKSAAN JARAK AMAN			
Jarak aman	m	min	OK
Flare -> tepi GH	15.8	15	Ya
Flare -> kontainer CHP	20.8	15	Ya
GH -> rumah boiler PKS	141.5	30	Ya
GH -> tangki CPO	208.0	30	Ya
Flare -> batas pagar lokasi	21.5	10	Ya
Reaktor -> kolam pendingin (tanah)	20.9	20	Ya
Cerobong boiler -> GH	137.1	30	Ya

CATATAN :
1. Arsir silang = kolam dibebaskan; arsir putus = air.
2. Efluen produk tetap ke kolam polishing sebelum aplikasi lahan.
3. Garis rantai = kabel 20 kV (PKS & PLN).

JALAN UMUM

250 m

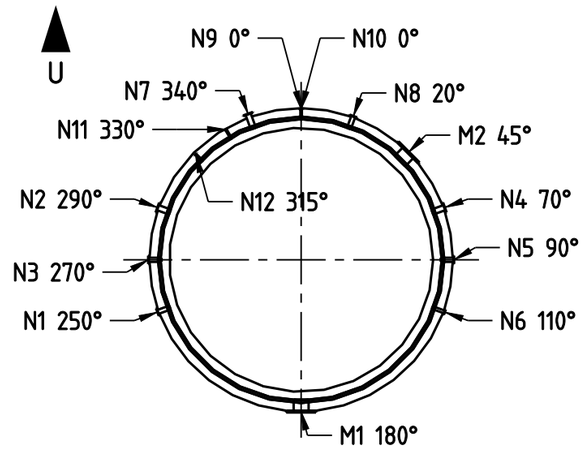
DENAH LOKASI (DARI MODEL 3D)

SKALA 1:1750

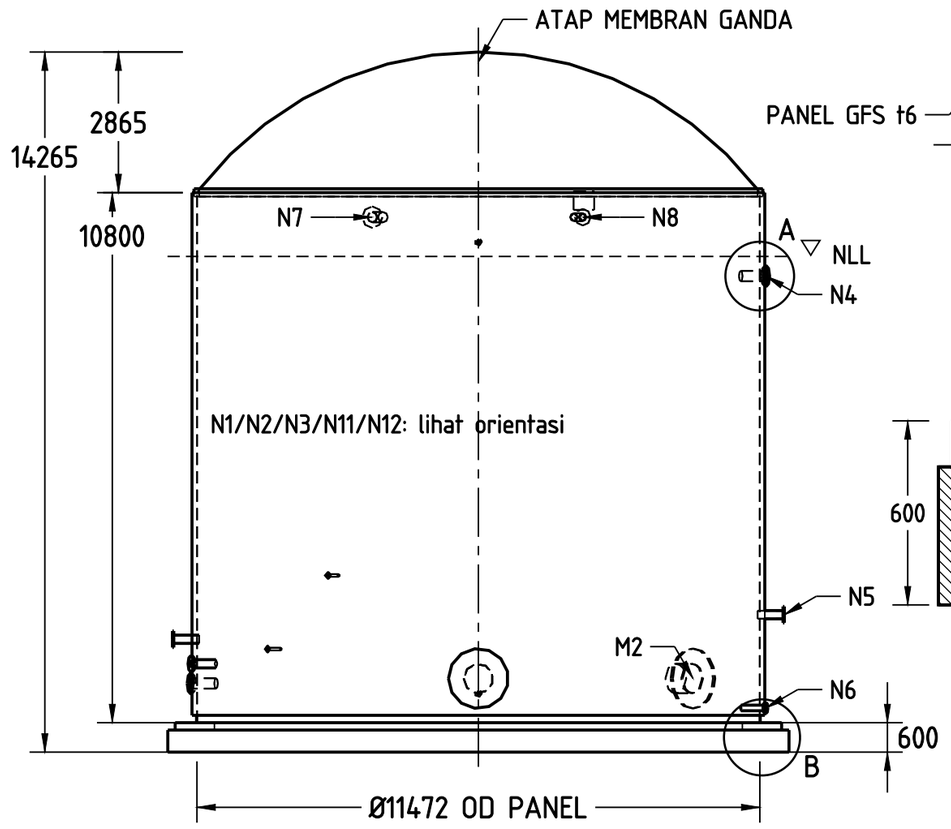
LEGENDA	
Kode	Uraian
WB	Jembatan timbang 2 x 60 t
OF	Kantor
LB	Laboratorium
WS	Bengkel
ST	Gudang
MS	Musala
LR	Loading ramp TBS 20 pintu
ST	Rebusan 3 x 30 m
MB	Pabrik utama
CL	Stasiun klarifikasi (CST, tangki minyak)
KS	Silo kernel
BH	Rumah boiler 2 x 30 t uap/jam
ER	Rumah mesin
RW	K. air baku
WT	Pengolahan air + menara air

Kode	Uraian
CT	Tangki CPO 2 x 2.000 t
EF	Hopper TKS
FP	Fat pit (sumber POME)
CP1	K. pendingin 1 (585 m³, 1 d)
CP2	K. pendingin 2 (585 m³, 1 d)
AC	K. pengasaman (2,925 m³, 5 d) (bebas)
AN1	K. anaerobik 1 (17,550 m³, 30 d) (bebas)
AN2	K. anaerobik 2 (17,550 m³, 30 d) (bebas)
FA	K. fakultatif (8,775 m³, 15 d)
AE1	K. aerobik 1 (4,388 m³, 7.5 d)
AE2	K. aerobik 2 (4,388 m³, 7.5 d)
LA	Flatbed aplikasi lahan (efluen ke kebun)
GI	Gardu step-up 0,4/20 kV 1,25 MVA
PRODUK	Set biogas kontak, plot 3757 m²

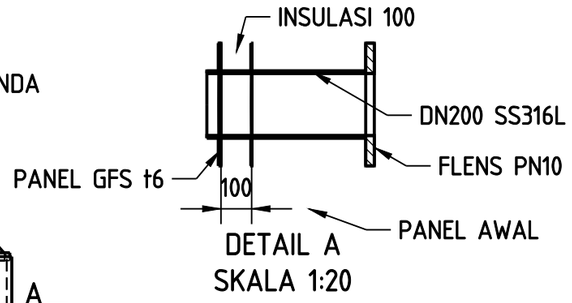
University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: Site Plan		Rev: 2
	Author : NAMA TIM	Denah Lokasi PKS 45 t TBS/jam + Set Biogas (Rev 2)	Unit: mm	Scale: 1 : 1750	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-GA-002		Sheet number: 9 / 9



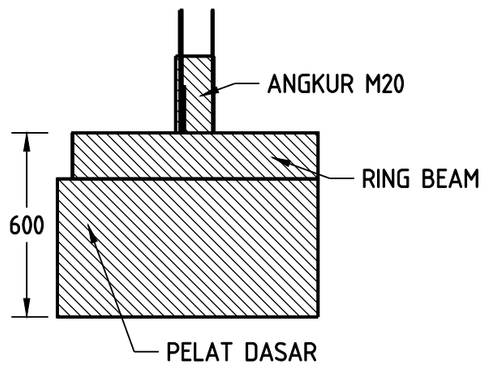
ORIENTASI NOSEL
SKALA 1:250



ELEVASI (DARI SELATAN)
SKALA 1:125



DETAIL A
SKALA 1:20



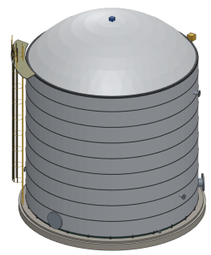
DETAIL B
SKALA 1:20

JADWAL NOSEL					
Tag	Ukuran	Rating	Layanan	Elev.	Orient.
N1	DN150	PN10 RF	Umpan reaktor (dari P-201)	1800	250°
N2	DN200	PN10 RF	Isap jet mixing (ke P-202)	1400	290°
N3	DN150	PN10 RF	Balik jet mixing (header nosel jet)	2300	270°
N4	DN200	PN10 RF	Keluaran ke degasser DG-301 (luapan)	9700	70°
N5	DN150	PN10 RF	Resirkulasi lumpur dari LS-301	2800	90°
N6	DN100	PN10 RF	Pengurasan dasar / pasir	900	110°
N7	DN200	PN10 RF	Keluaran biogas (ruang gas)	10900	340°
N8	DN150	PN10 RF	PVRV tekanan-vakum	10900	20°
N9	DN50	PN10 RF	Pemancar level LT atas	10400	0°
N10	DN50	PN10 RF	Pemancar level LT bawah	1200	0°
N11	DN50	PN10 RF	Termometer TT / pH	3600	330°
N12	DN50	PN10 RF	Titik sampel	2100	315°
M1	DN600	PN10 RF	Lubang orang dinding (bawah)	1500	180°
M2	DN600	PN10 RF	Lubang orang dinding (seberang)	1500	45°

DATA DESAIN	
Data desain	Nilai
Fungsi / jumlah	Reaktor kontak termofilik, 3 unit identik (2 / 4 untuk 30 / 60 t/jam)
Kode desain	AWWA D103-09 (tangki baja dibaut) + ISO 28765 (enamel)
Tekanan / suhu desain	+50 / −5 mbar(g) + hidrostatik penuh; 70 °C (op. 55 °C)
Volume dinding / kerja	1114 m³ / 980 m³, NLL 9500 mm dari lantai
Panel	126 panel 2700 x 1200 mm, t 5–10 mm (rata-rata 6), enamel 2 lapis
Atap	Membran ganda PVC-PES (dalam: gas, luar: udara), tinggi 2865 mm
Angin / gempa	SNI 1727:2020 (40 m/s) / SNI 1726:2019, Kelas D
Insulasi / kladding	Mineral wool 100 mm 100 kg/m³ / Al 0,8 mm + sabuk
Massa baja / operasi	± 20.3 t / ± 1020 t
Pengiriman / relokasi	Panel datar dalam 2 kontainer 40 ft; ereksi ± 3 minggu; dapat dibongkar

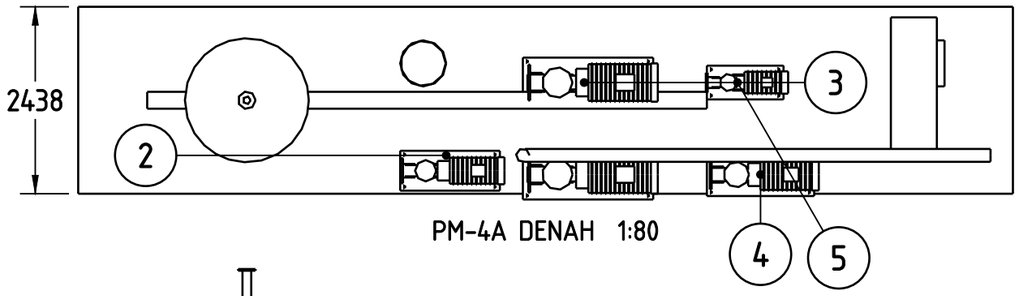
DAFTAR MATERIAL (MASSA DARI MODEL 3D)				
No	Bagian	Material	Jml	Massa kg
1	Panel dinding GFS	S355 + enamel 2 lapis	126	18663
2	Baut panel + sealant	M12 galv. + mastik	± 7560	–
3	Sudut penahan + angkur M20	S275 galv.	24	452
4	Leher nosel + flens	SS316L / SA-105	14	1667
5	Header jet internal + 8 nosel	SS316L DN150 sch10	25.6 m	282
6	Atap membran ganda + klem	PVC-PES 1.100 g/m² / Al	1	± 950
7	Blower udara membran luar	0,75 kW + katup cek	1	60
8	Tangga + sangkar + platform	S275 galv., grating	1	1267
9	Insulasi mineral wool	100 kg/m³	39.3 m³	3926
10	Kladding + sabuk	Al 3003 t0,8	1	1203
11	Ring beam + pelat dasar	Beton K-300 (sipil lokal)	1	–

CATATAN :
1. Panel dibaut dengan sealant; tanpa las di lokasi.
2. Uji: isi air 24 jam (kebocoran), uji udara atap 5 mbar.
3. Elevasi dari muka tanah (EL 0); orientasi dari utara.
4. Relokasi: bongkar panel, ganti sealant & ring beam.

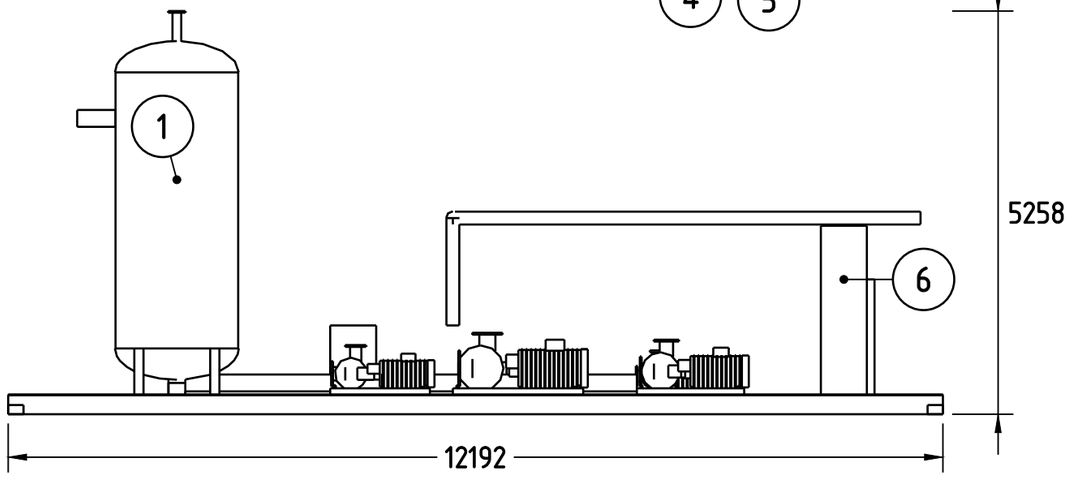


ISOMETRI

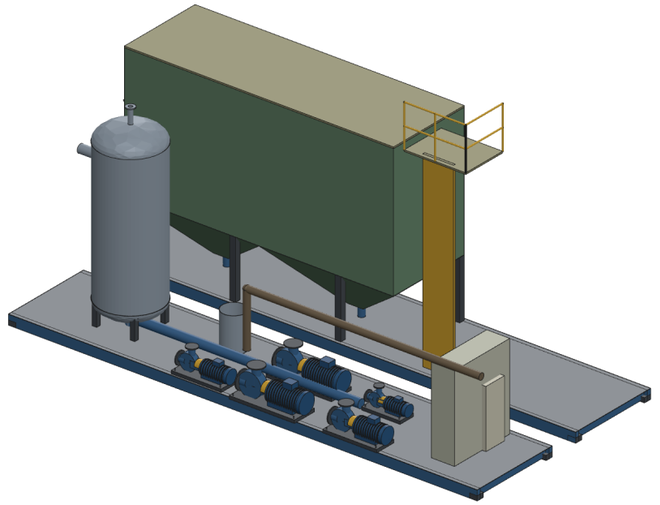
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Equipment Drawing	Rev: 2
	Author : NAMA TIM	Reaktor Kontak Termofilik R-201 s/d R-203 (GFS)	Unit: mm / Scale: 1:125/1:250/1:20 / Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-V-201	Sheet number: 2 / 9



PM-4A DENAH 1:80



PM-4A TAMPAK SAMPING 1:80

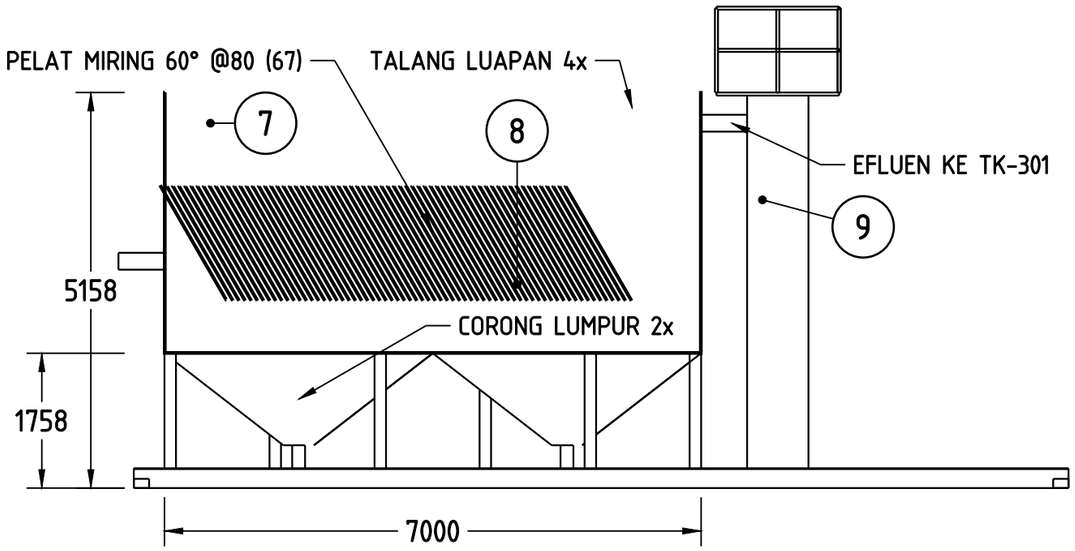


ISOMETRI PM-4A + PM-4B (TANPA SKALA)

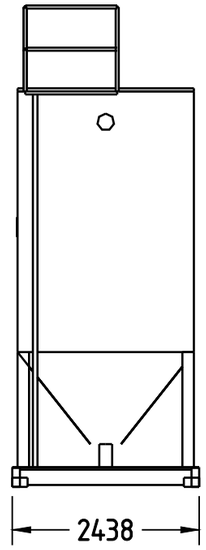
DAFTAR BAGIAN					
No	Tag	Uraian	Bahan	Jml	kg
1	DG-301	Degasser vakum Ø1600 x 4000, 8 m³	SA-516-70 / epoksi	1	1850
2	K-303	Pompa vakum cincin cair + separator	CF8M, 5,5 kW	1	420
3	P-301A/B	Pompa resirkulasi lumpur 585 m³/hari	CF8M, 7,5 kW, VSD	2	2x560
4	P-302	Pompa umpan settler	CF8M, 5,5 kW	1	430
5	P-303	Pompa lumpur buangan -> PM-10	PC, 3 kW	1	260
6	–	Panel MCC + PLC + analis FOS/TAC	IP54	1	820
7	LS-301	Settler lamella 81 m² proyeksi	S275 + epoksi	1	6900
8	–	Paket pelat miring 60°, jarak 80 mm	PP / FRP	1	1300
9	–	Tangga + platform akses	S275 galv.	1	540
10	–	Skid 40 ft x 2 + perpipaan, katup	S275 galv.	2	2x2600

CATATAN :

1. Proses kontak: reaktor -> DG-301 (vakum -0,3 bar, CH4 terlarut kembali ke jalur gas) -> LS-301 -> underflow 585 m³/hari kembali ke reaktor (P-301), buangan 110 m³/hari ke screw press (P-303).
2. Tangkapan padatan 85 %; SRT 12 hari.
3. Masuk LS-301 dari DG-301 di ujung barat (kiri B-B).
4. Membran AnMBR = opsi PM-11 (tidak termasuk).

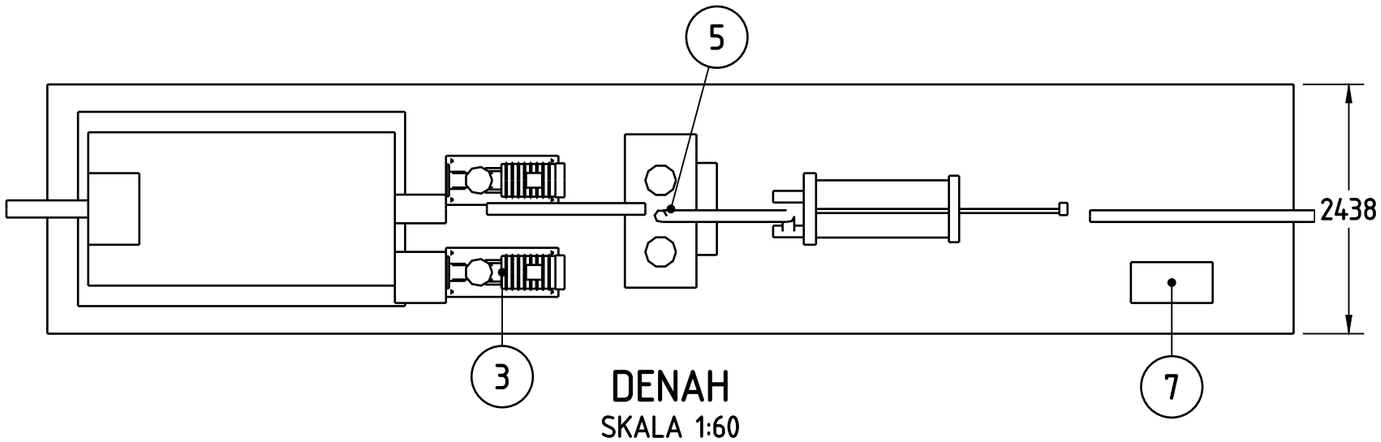


PM-4B POTONGAN B-B (SETTLER LAMELLA) 1:80

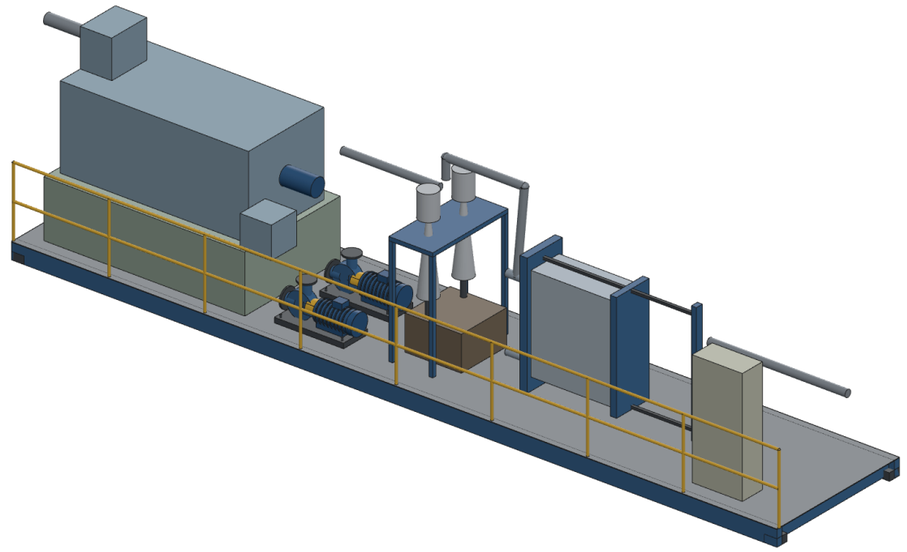


PM-4B UJUNG 1:80

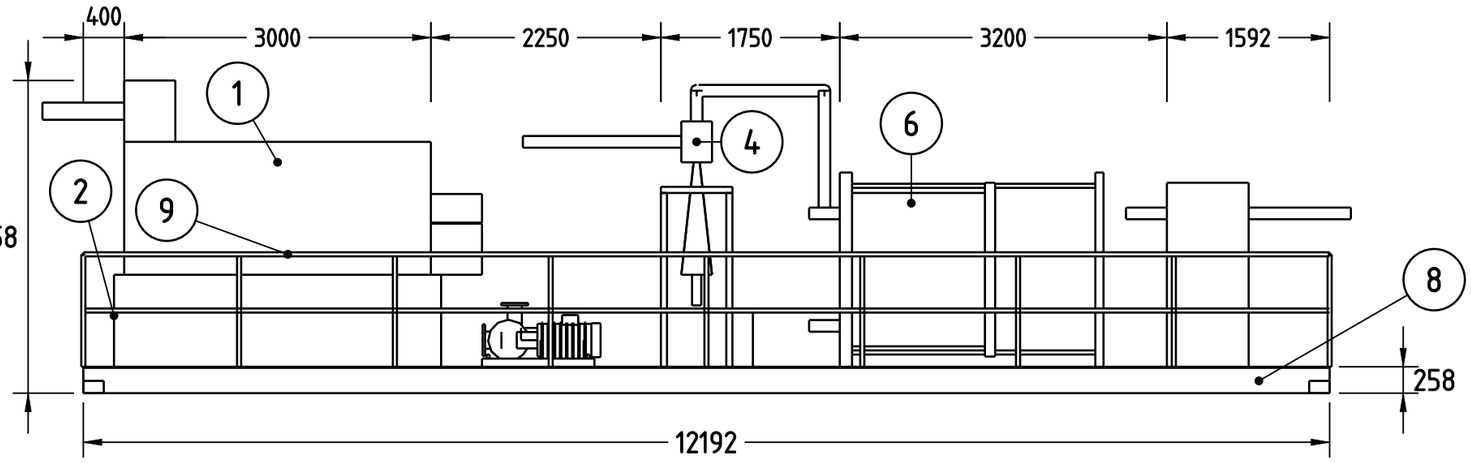
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing	Rev: 2
	Author : NAMA TIM	PM-4 Skid Kontak DG-301 + LS-301	Unit: mm / Scale: 1 : 80 / Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-004	Sheet number: 3 / 9



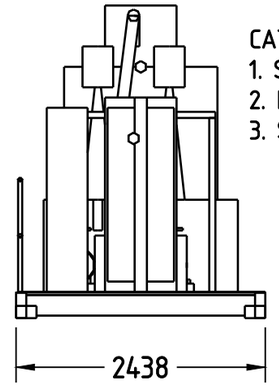
DENAH
SKALA 1:60



ISOMETRI (TANPA SKALA)



TAMPAK DEPAN
SKALA 1:60



TAMPAK UJUNG
SKALA 1:60

CATATAN :
1. Seluruh alat dirakit & diuji di bengkel.
2. Pengiriman: 1 truk trailer 40 ft, ± 12 t.
3. Sambungan: POME masuk DN150 (kiri), POME keluar DN100 ke TK-101 (kanan).

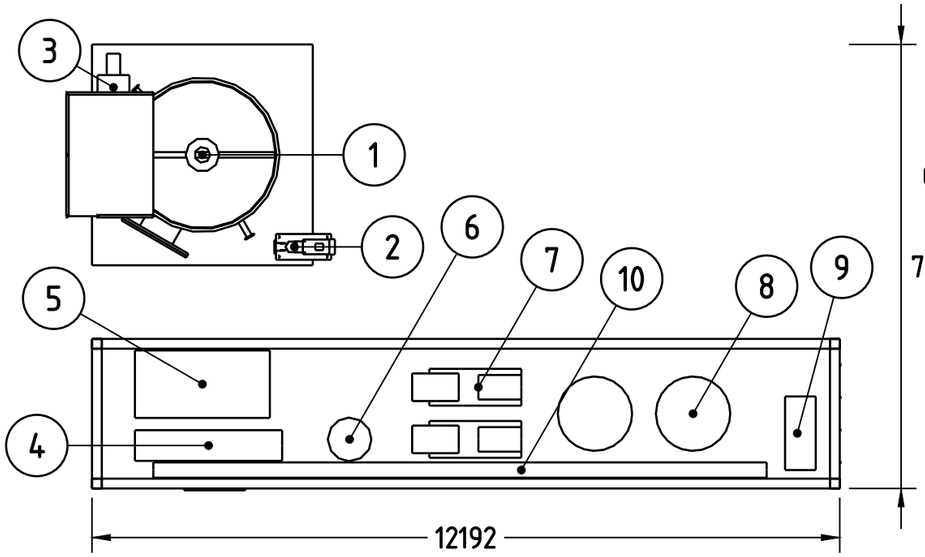
DAFTAR BAGIAN

No	Tag	Uraian	Bahan	Jml	kg
1	SC-101	Saringan drum Ø900x2600, 0,75 mm	SS304	1	1450
2	–	Bak sump 3 m³	SS304 t4	1	620
3	P-101A/B	Pompa desander H 20 m, VSD	CD4MCu, 11 kW	2	2x310
4	HC-101	Hidrosiklon Ø300	PU-lined CS	2	2x140
5	–	Bak pasir 0,4 m³	S275 galv.	1	180
6	E-101	PHE 827 kWth	SS316/EPDM	1	1850
7	–	Panel kontrol lokal	IP65	1	220
8	–	Skid UNP 250 12192x2438	S275, cat	1	3900
9	–	Pegangan tangan Ø42,4	S275 galv.	1	240
10	–	Pipa, katup, kabel	SS304 / PVC	1 lot	1100

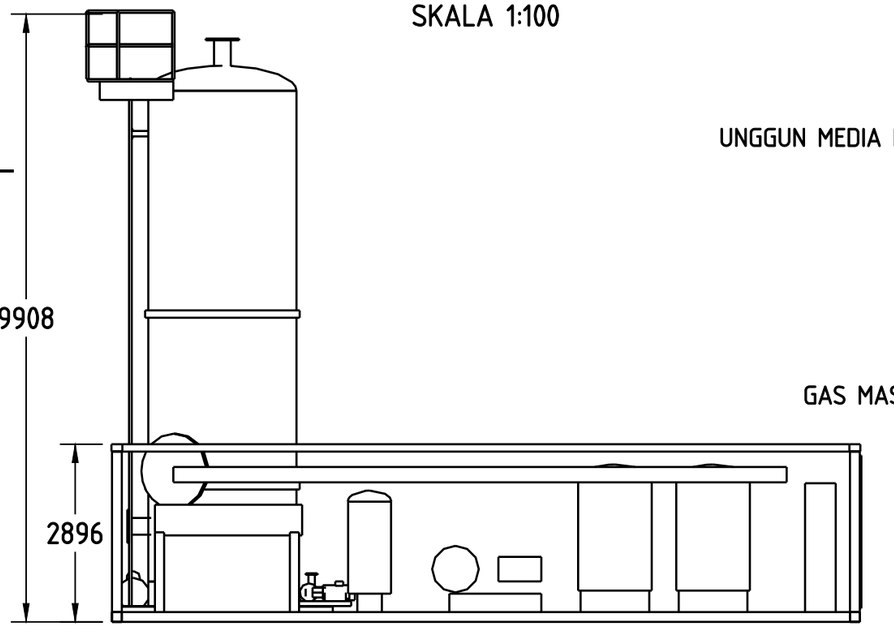
SAMBUNGAN SKID

Tag	Ukuran	Layanan	Dari → ke
N-A	DN150	POME panas masuk (85 °C)	PKS → SC-101
N-B	DN100	POME tersaring ke TK-101	E-101 → TK-101
N-C	DN80	Air pendingin masuk / keluar	CT-601 ↔ E-101
N-D	DN50	Underflow pasir	HC-101 → bak
N-E	–	Daya 400 V, 3 fasa, 45 kVA	MCC → panel

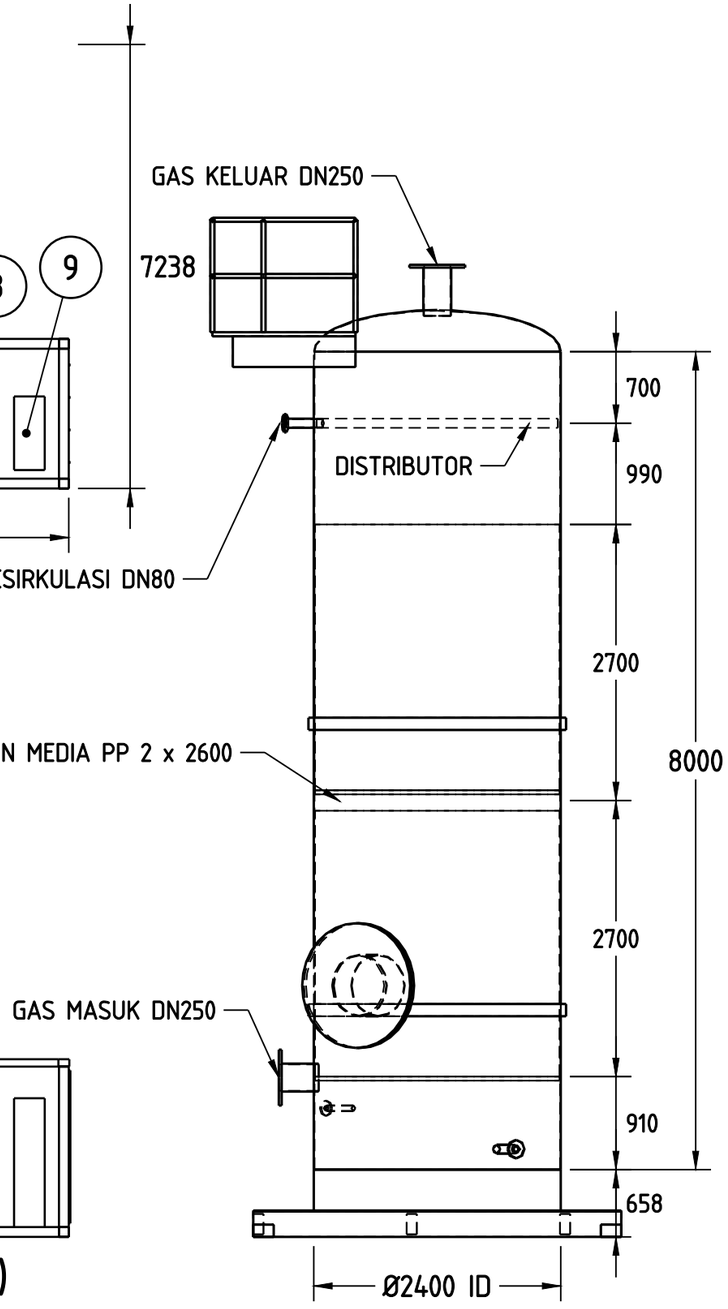
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing	Rev: 0
	Author : NAMA TIM	PM-1 Skid Praolah POME (SC-101, HC-101, E-101)	Unit: mm / Scale: 1 : 60 / Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-001	Sheet number: 4 / 9



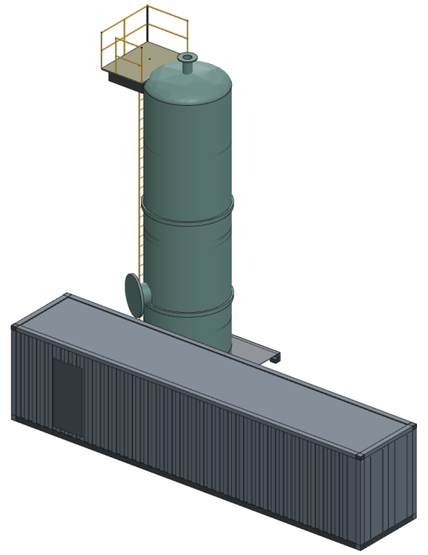
DENAH (ATAP DILEPAS)
SKALA 1:100



TAMPAK DEPAN (DINDING DILEPAS)
SKALA 1:100



BTF-301 (ELEVASI)
SKALA 1:60



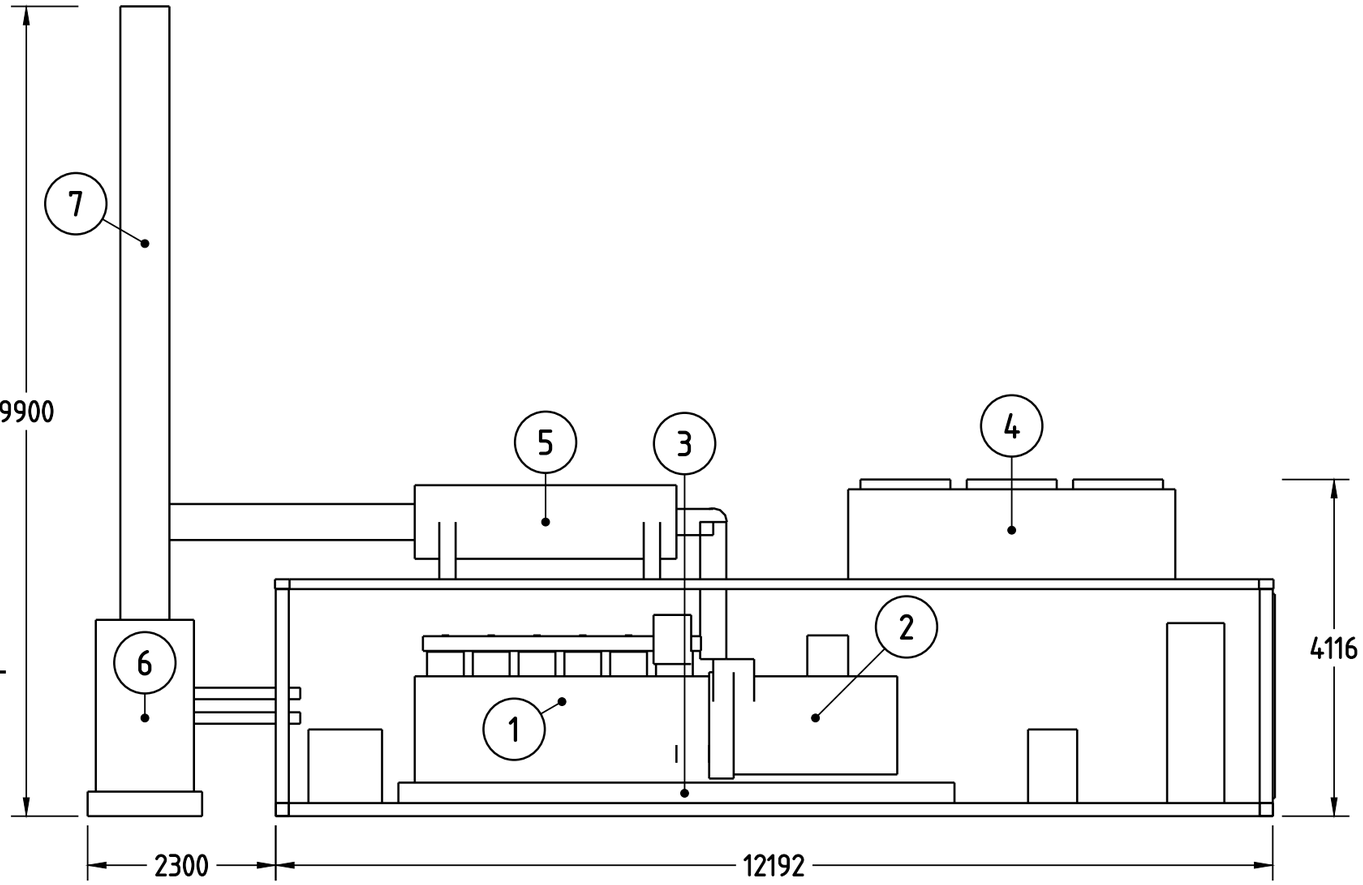
ISOMETRI (TANPA SKALA)

DAFTAR BAGIAN					
No	Tag	Uraian	Bahan	Jml	kg
1	BTF-301	Bio-trickling filter Ø2400x8000	FRP vinilester	1	3800
2	P-305	Pompa resirkulasi nutrien	PP, 1,5 kW	1	60
3	K-301	Blower udara BTF	Al, 0,75 kW	1	45
4	HX-301	Pendingin gas shell-tube	SS316L	1	650
5	CH-301	Chiller paket	R-32, 15 kWe	1	780
6	V-301	KO pot Ø700 + demister	SS304	1	210
7	K-302A/B	Blower biogas 538 m³/jam	Al, ATEX 18,5 kW	2	2x420
8	AC-301A/B	Filter karbon aktif Ø1200	CS epoksi + KI-AC	2	2x1650
9	–	Panel kontrol + analis gas	IP54, ATEX	1	380
10	–	Header gas DN250	SS304	1	420
11	–	Kontainer 40 ft HC, ATEX z2	Corten	1	4300
12	–	Skid BTF 3600 + tangga	S275 galv.	1	1650

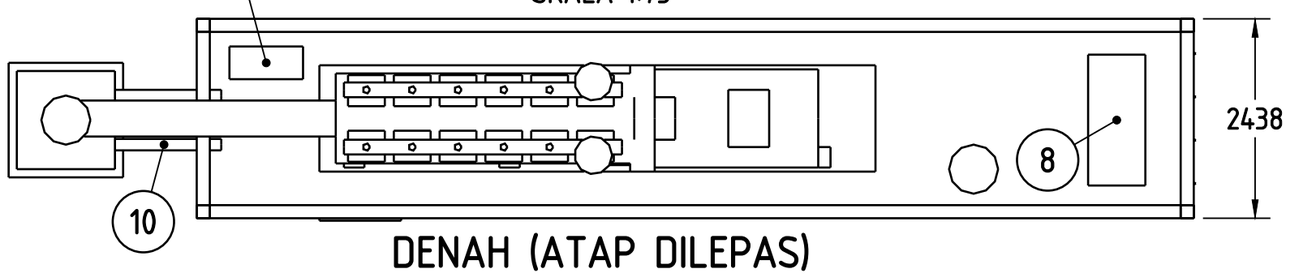
NOSEL BTF-301		
Tag	Ukuran	Layanan BTF-301
G1	DN250	Gas masuk (bawah)
G2	DN250	Gas keluar (puncak)
A1	DN50	Udara dari K-301
L1	DN80	Resirkulasi ke distributor
L2	DN80	Sump / blowdown
MH	DN600	Lubang orang

CATATAN :
1. BTF dikirim terpisah di trailer.
2. H_2S 2868 → 21 ppm.
3. Kontainer: ventilasi paksa + detektor gas.

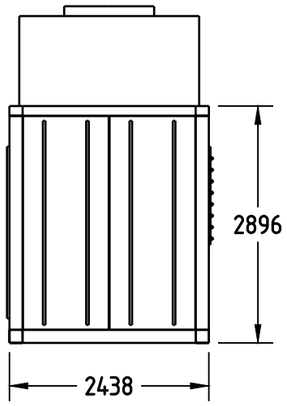
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	PM-6 Skid Pengolahan Biogas (BTF-301, K-302, AC-301)	Unit: mm	Scale: 1:100 / 1:60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-006			Sheet number: 5 / 9



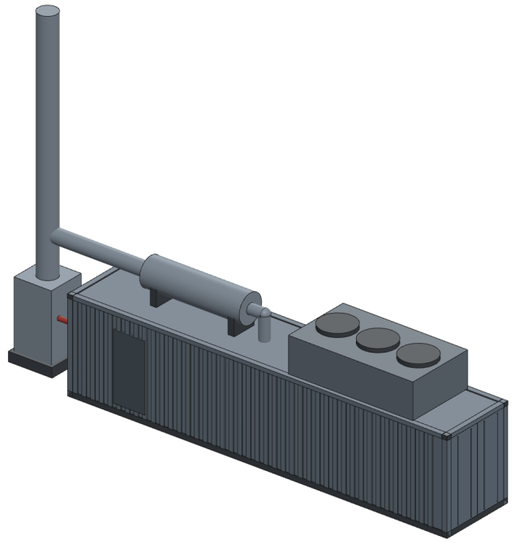
TAMPAK DEPAN (DINDING DILEPAS)
SKALA 1:75



DENAH (ATAP DILEPAS)
SKALA 1:75



TAMPAK UJUNG
SKALA 1:75



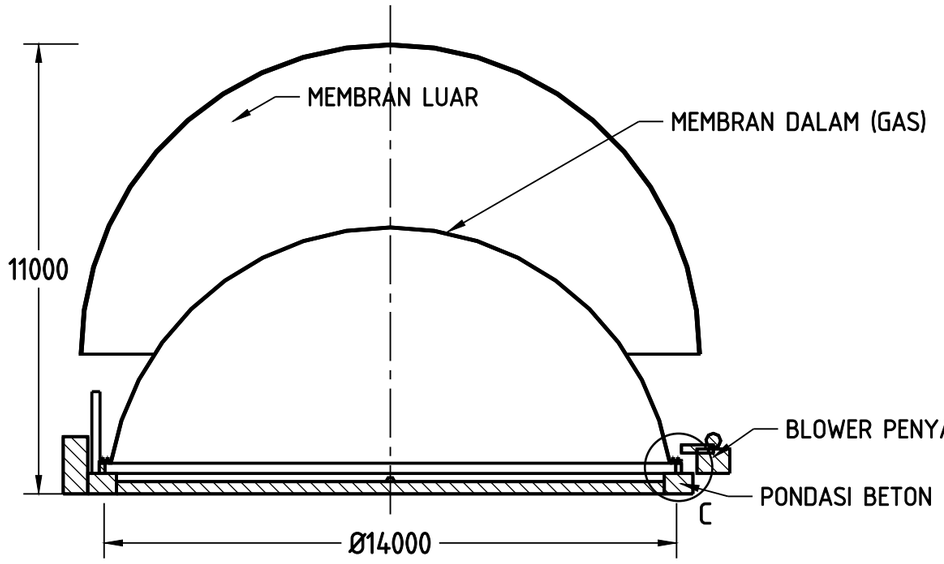
ISOMETRI (TANPA SKALA)

DAFTAR BAGIAN (PER UNIT)					
No	Tag	Uraian	Bahan	Jml	kg
1	GE-401	Mesin gas V-12, 600 kWe	blok besi cor	1	7800
2	–	Generator sinkron 400 V 50 Hz	IP23, H	1	3100
3	–	Rangka dasar + peredam	S355 + karet	1	1400
4	–	Radiator / dump cooler	Al/Cu, 3 kipas	1	1250
5	–	Silencer Ø900x3200	CS, 35 dB(A)	1	650
6	HX-401	Pemulih panas gas buang	SS316L/CS	1	1900
7	ST-401	Cerobong Ø600 x 9,9 m	CS galv.	1	850
8	–	Panel kontrol + sinkron	IP54, 1600 A	1	900
9	–	Gas train + regulator	DN100, ATEX	1	260
10	–	Pipa HWS/HWR DN80	CS Sch40 insul.	2	2x60
11	–	Kontainer 40 ft HC, 75 dB(A)	Corten + insul.	1	5200

DATA KINERJA GE-401	
Data genset	Nilai
Daya listrik terpasang / operasi	600 kWe / 563 kWe (beban 94 %)
Efisiensi listrik (beban parsial)	39.3 %
Bahan bakar per unit	1433 kW (LHV)
Panas pulih (jaket + gas buang)	703 kWth
Suhu gas buang masuk HX / cerobong	460 °C / 150 °C
Emisi (Permen LHK 11/2021)	NOx ≤ 500, CO ≤ 1000 mg/Nm³

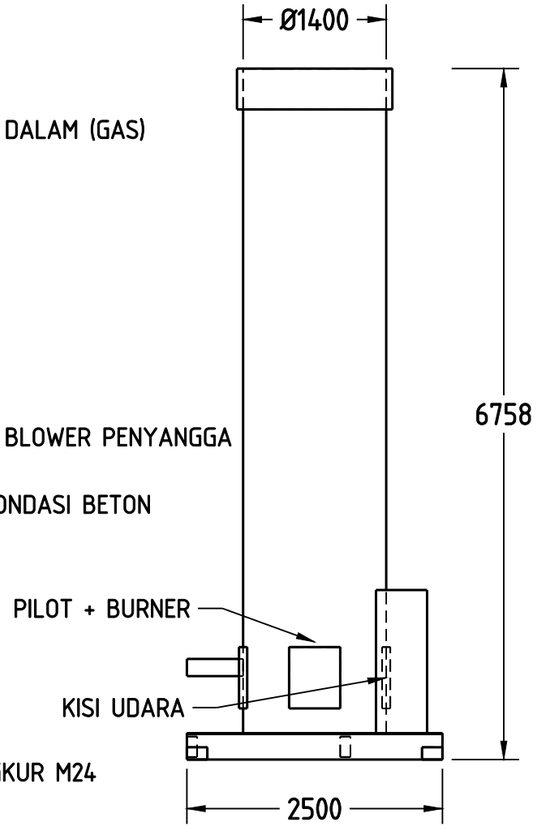
CATATAN :
1. Dua unit identik GE-401 / GE-402 (2 kontainer).
2. Bahan bakar: biogas H_2S ≤ 50 ppm dari PM-6.
3. HX-401 & ST-401 dikirim sebagai skid PM-9.
4. Massa angkut ± 24 t per kontainer.

University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing	Rev: 0
	Author : NAMA TIM	PM-8 Kontainer CHP 600 kWe + HX-401 / ST-401 (PM-9)	Unit: mm / Scale: 1 : 75 / Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-008	Sheet number: 6 / 9



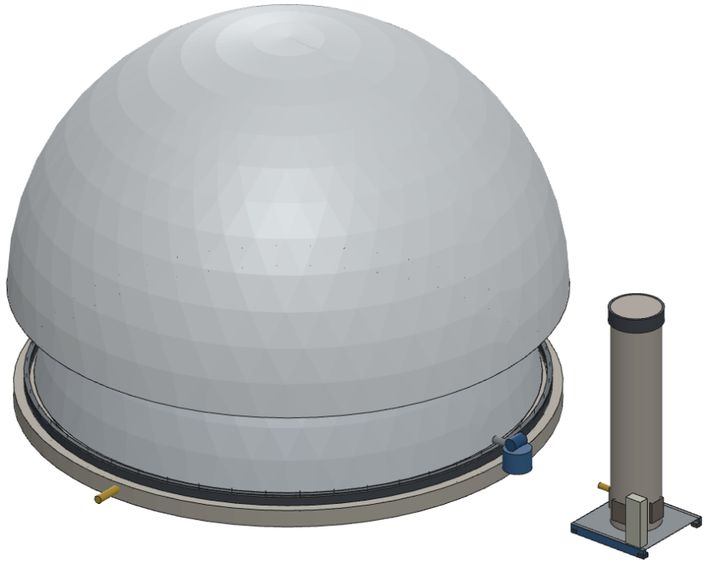
POTONGAN A-A GH-301

SKALA 1:150



FL-301 (ELEVASI)

SKALA 1:60

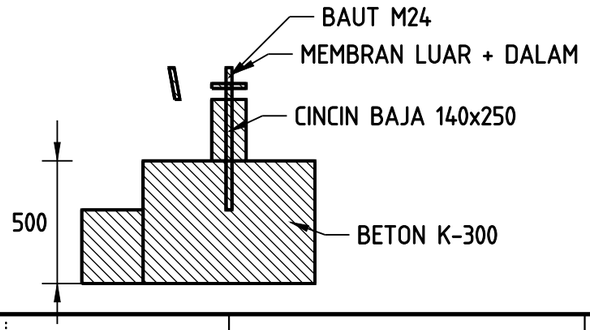


ISOMETRI (TANPA SKALA)

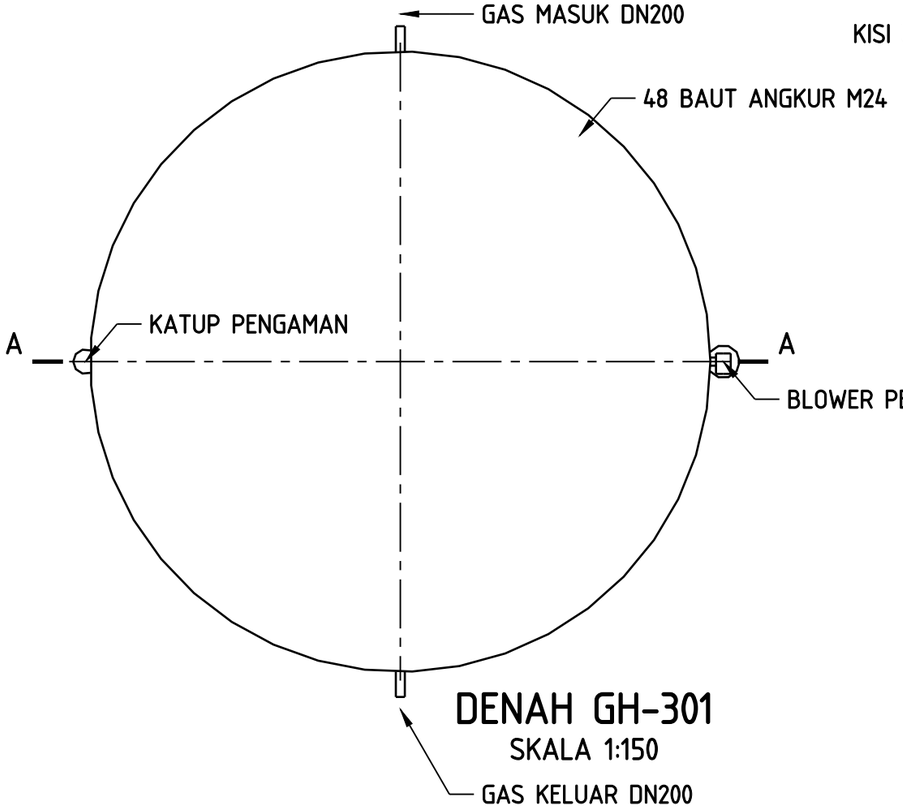
DATA DESAIN		
Tag	Parameter	Nilai
GH-301	Volume gas berguna	986 m³ (± 2 jam P95)
	Volume kubah total	1414 m³, Ø14.0 x 10.5 m
	Membran luar / dalam	PVC-poliester 1100 g/m² / PE kedap metana
	Tekanan operasi	2 – 3 mbar(g), blower udara penyangga
	Pondasi	Cincin beton Ø14800, lantai beton 300 mm
	Pengiriman	membran terlipat di 1 kontainer 20 ft
FL-301	Kapasitas	542 Nm³/jam, suhu nyala ≥ 1000 °C
	Tipe	Flare tertutup, pilot otomatis, skid 2,5 x 2,5 m

CATATAN :

1. Membran luar dijaga tegang oleh blower udara penyangga.
2. Membran dalam naik-turun sesuai isi gas (LIC-401).
3. Flare menyala otomatis saat GE berhenti / GH penuh (PIC-407).
4. Jarak aman GH ke flare ≥ 15 m (lihat tata letak BG-GA-001).



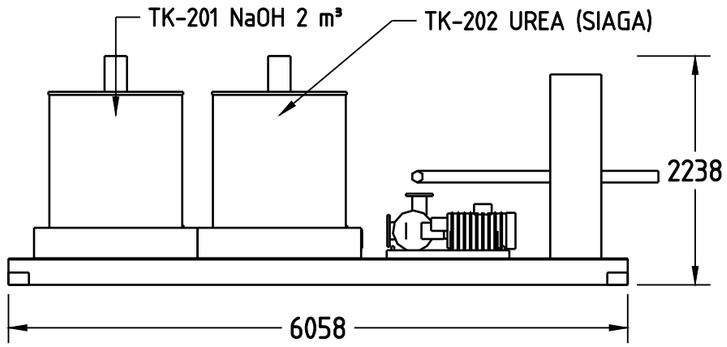
DETAIL C
SKALA 1:25



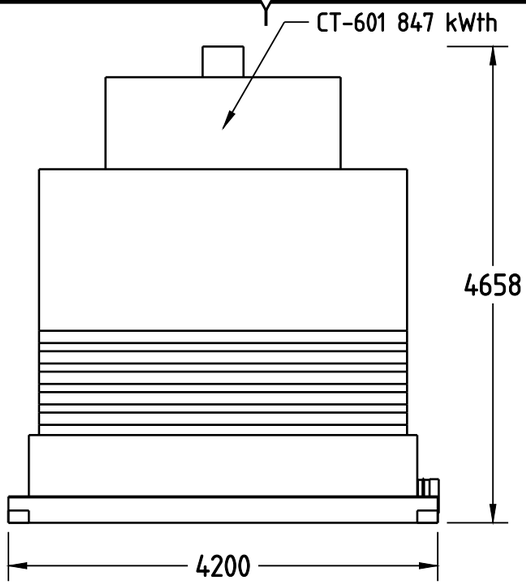
DENAH GH-301

SKALA 1:150

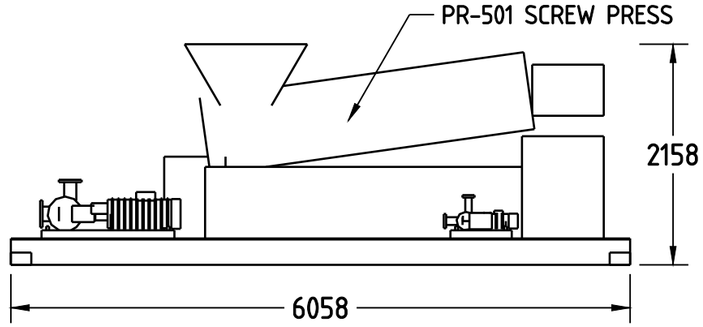
University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: Assembly Drawing		Rev: 0
	Author : NAMA TIM	PM-7 Penampung Gas GH-301 + Flare Tertutup FL-301	Unit: mm	Scale: 1:150 / 1:60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-M-007		Sheet number: 7 / 9



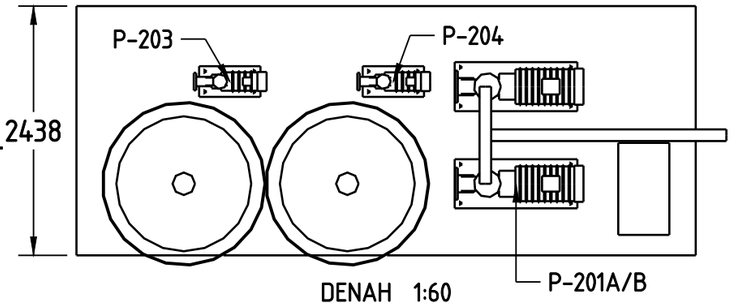
PM-5 TAMPAK DEPAN 1:60



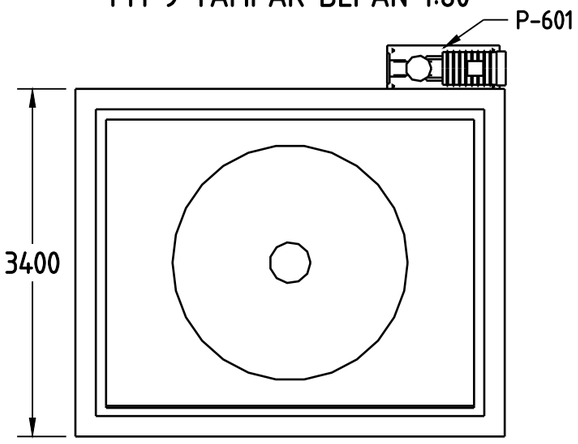
PM-9 TAMPAK DEPAN 1:60



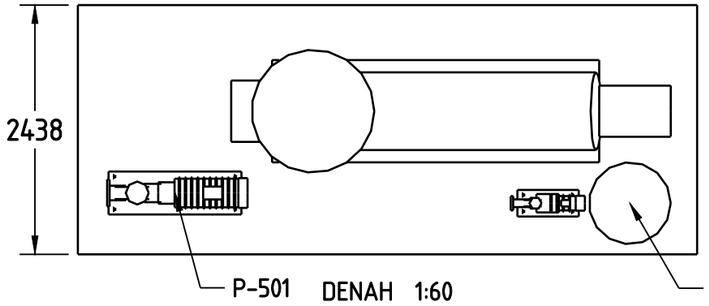
PM-10 TAMPAK DEPAN 1:60



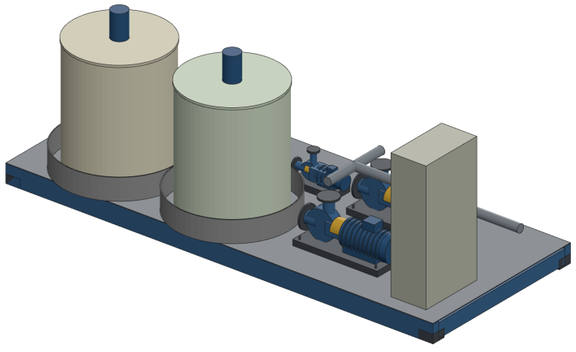
DENAH 1:60



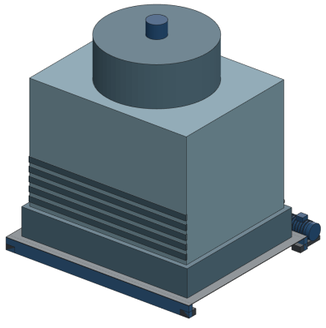
DENAH 1:60



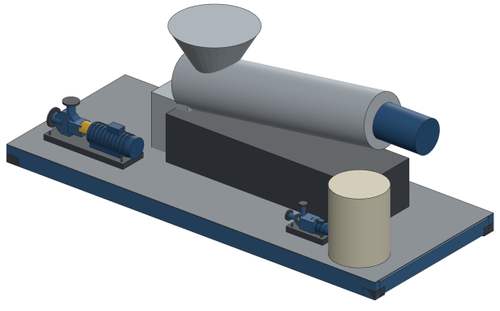
DENAH 1:60



ISOMETRI PM-5



ISOMETRI PM-9



ISOMETRI PM-10

DAFTAR BAGIAN SKID				
Skid	Tag	Uraian	Bahan / daya	kg
PM-5	TK-201/202	Tangki NaOH / urea (siaga) 2 m³	HDPE + bund	2x260
	P-201A/B	Pompa umpan reaktor	CF8M, 7,5 kW	2x260
	P-203/204	Pompa dosing	PVDF, 0,37 kW	2x45
	-	Panel dosing + skid 20 ft	IP65 / S275	2080
PM-9	CT-601	Menara pendingin 847 kWth	FRP, 7,5 kW	2900
	P-601	Pompa air pendingin	GG25, 11 kW	210
	-	Skid 4200 x 3400	S275 galv.	1500
PM-10	P-501	Pompa umpan press (PC)	CS/NBR, 4 kW	380
	PR-501	Screw press lumpur	SS304, 5,5 kW	2600
	TK/P-polimer	Tangki + pompa polimer 1 m³	HDPE / PVDF	160
	-	Skid 20 ft + chute kue	S275 galv.	1900

University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	Skid PM-5, PM-9 dan PM-10 (Dosing, CT-601, Lumpur)	Unit: mm	Scale: 1 : 60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-M-005			Sheet number: 8 / 9